

## HOMEWORK 1

1. Compute the volume of the parallelepiped given below. The edges of the parallelepiped are formed the three vectors which are,

$$\mathbf{u} = 3\mathbf{e}_1 - \mathbf{e}_2$$

$$\mathbf{v} = \mathbf{e}_2 - 2\mathbf{e}_3$$

$$\mathbf{w} = \mathbf{e}_1 + 5\mathbf{e}_2 + 4\mathbf{e}_3$$

2. Interpret  $\text{grad } \phi$ ,  $\text{div } \mathbf{A}$ , and  $\nabla^2 \phi$  in both cylindrical and spherical coordinates.

3. If  $\alpha \neq 0$  and  $\phi(\alpha) = \int_{\alpha}^{\alpha^2} \frac{\sin \alpha x}{x} dx$  then compute  $\phi'(\alpha)$  using the Leibnitz Theorem.

4. If  $\mathbf{r}$  is given as  $\mathbf{r} = (x^2 \sin y, z^2 \cos y, -xy^2)$  the compute  $d\mathbf{r}$ .

5.  $\phi = x^2 y z^3$ ,  $\mathbf{A} = (xz, -y^2, 2x^2 y)$  are given in Cartesian coordinate system, compute;

i.  $\nabla \phi$

ii.  $\nabla \cdot \mathbf{A}$

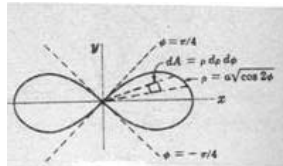
iii.  $\text{div}(\phi \mathbf{A})$

iv.  $\text{rot}(\phi \mathbf{A})$

6. If  $\int_0^{\pi} \frac{dx}{\alpha - \cos x} = \frac{\pi}{\sqrt{\alpha^2 - 1}}$  and  $\alpha > 1$  are given, using Leibnitz's theorem compute

$$\int_0^{\pi} \frac{dx}{(2 - \cos x)^2} \quad (\text{Do not use direct integration!})$$

7. Compute the area of the region bounded by the lemniscate's curve  $\rho = a^2 \cos(2\phi)$  on the xy plane.



Aslı TÜRKER

030000261

Tekstil Mühendisliği

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FLUID MECHANICS

~ HOMEWORK I ~

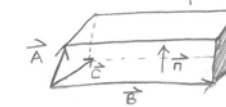
Compute the volume of the parallelepiped given below. The edges of the parallelepiped are formed the three vectors which are

$$\mathbf{u} = 3\vec{e}_1 - \vec{e}_2 \quad \mathbf{v} = \vec{e}_2 - 2\vec{e}_3 \quad \mathbf{w} = \vec{e}_1 + 5\vec{e}_2 + 4\vec{e}_3$$

$$\vec{e}_1 = (1, 0, 0) \quad \vec{e}_2 = (0, 1, 0) \quad \vec{e}_3 = (0, 0, 1) \Rightarrow$$

$$\mathbf{u} = 3\vec{i} - \vec{j} \quad \mathbf{v} = \vec{j} - 2\vec{k} \quad \mathbf{w} = \vec{i} + 5\vec{j} + 4\vec{k}$$

Volume of the parallelepiped = (h). (area of an parallelogram)



$$= (\vec{A}, \vec{n}) (|\vec{B} \times \vec{C}|)$$

$$= \vec{A} \cdot \{ |\vec{B} \times \vec{C}| \vec{n} \} = \vec{A} \cdot (\vec{B} \times \vec{C})$$

$$|\vec{A} \cdot (\vec{B} \times \vec{C})| = \begin{vmatrix} 3 & -1 & 0 \\ 0 & 1 & -2 \\ 1 & 5 & 4 \end{vmatrix} = [3(4+10)] + [0 \cdot (0-1)] - [(-1) \cdot (0+2)]$$

$$= |42+2| = |44| = 44$$

2. Interpret  $\text{grad } \Phi$ ,  $\text{div } \mathbf{A}$ , and  $\nabla^2 \Phi$  in both cylindrical and spherical coordinates.

in cylindrical coordinates;

$$i) \nabla \Phi = \text{grad } \Phi = \frac{1}{h_1} \frac{\partial \Phi}{\partial u_1} \vec{e}_1 + \frac{1}{h_2} \frac{\partial \Phi}{\partial u_2} \vec{e}_2 + \frac{1}{h_3} \frac{\partial \Phi}{\partial u_3} \vec{e}_3$$

$$u_1 = \rho \quad u_2 = \phi \quad u_3 = z \quad h_1 = 1 \quad h_2 = \rho \quad h_3 = 1$$

$$= \frac{1}{1} \frac{\partial \Phi}{\partial \rho} \vec{e}_1 + \frac{1}{\rho} \frac{\partial \Phi}{\partial \phi} \vec{e}_2 + \frac{1}{1} \frac{\partial \Phi}{\partial z} \vec{e}_3$$

$$= \frac{\partial \Phi}{\partial \rho} \vec{e}_1 + \frac{1}{\rho} \frac{\partial \Phi}{\partial \phi} \vec{e}_2 + \frac{\partial \Phi}{\partial z} \vec{e}_3$$

$$ii) \nabla \cdot \vec{A} = \text{div } \vec{A} = \frac{1}{h_1 h_2 h_3} \left[ \frac{\partial}{\partial u_1} (h_2 h_3 A_1) + \frac{\partial}{\partial u_2} (h_3 h_1 A_2) + \frac{\partial}{\partial u_3} (h_1 h_2 A_3) \right]$$

$$\begin{aligned} \nabla \cdot \vec{A} &= \frac{1}{(1)(\rho)(1)} \left[ \frac{\partial}{\partial \rho} (\rho(1) A_1) + \frac{\partial}{\partial \phi} (1(1) A_2) + \frac{\partial}{\partial z} (1(\rho) A_3) \right] \\ &= \frac{1}{\rho} \left[ \frac{\partial}{\partial \rho} (\rho A_1) + \frac{\partial A_2}{\partial \phi} + \frac{\partial A_3}{\partial z} \right] \\ (\vec{A} &= A_1 \vec{e}_1 + A_2 \vec{e}_2 + A_3 \vec{e}_3) \end{aligned}$$

$$\begin{aligned} iii) \nabla^2 \Phi &= \frac{1}{h_1 h_2 h_3} \left[ \frac{\partial}{\partial u_1} \left( \frac{h_2 h_3}{h_1} \frac{\partial \Phi}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left( \frac{h_3 h_1}{h_2} \frac{\partial \Phi}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left( \frac{h_1 h_2}{h_3} \frac{\partial \Phi}{\partial u_3} \right) \right] \\ &= \frac{1}{(1)(\rho)(1)} \left[ \frac{\partial}{\partial \rho} \left( \frac{(\rho)(1)}{(1)} \frac{\partial \Phi}{\partial \rho} \right) + \frac{\partial}{\partial \phi} \left( \frac{(1)(1)}{\rho} \frac{\partial \Phi}{\partial \phi} \right) + \frac{\partial}{\partial z} \left( \frac{(1)(\rho)}{(1)} \frac{\partial \Phi}{\partial z} \right) \right] \\ &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left( \rho \frac{\partial \Phi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{\partial^2 \Phi}{\partial z^2} \end{aligned}$$

in spherical coordinates;

$$i) \text{grad } \Phi = \frac{\partial \Phi}{\partial r} \vec{e}_1 + \frac{1}{r} \frac{\partial \Phi}{\partial \theta} \vec{e}_2 + \frac{1}{r \sin \theta} \frac{\partial \Phi}{\partial \phi} \vec{e}_3$$

$$ii) \text{div } \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_1) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_2) + \frac{1}{r \sin \theta} \frac{\partial A_3}{\partial \phi}$$

$$(\vec{A} = A_1 \vec{e}_1 + A_2 \vec{e}_2 + A_3 \vec{e}_3)$$

$$iii) \nabla^2 \Phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Phi}{\partial \phi^2}$$

3. If  $\alpha \neq 0$  and  $\phi(\alpha) = \int_{\alpha}^{\alpha^2} \frac{\sin \alpha x}{x} dx$  then compute  $\phi'(\alpha)$  using the Leibnitz Theorem.  $\alpha$

Using Leibnitz's Theorem;

$$\begin{aligned} \phi'(\alpha) &= \int_{\alpha}^{\alpha^2} \frac{\partial}{\partial \alpha} \left( \frac{\sin \alpha x}{x} \right) dx + \frac{\sin(\alpha \cdot \alpha^2)}{\alpha^2} \frac{d}{d\alpha} (\alpha^2) \\ &\quad - \frac{\sin(\alpha \cdot \alpha)}{\alpha} \frac{d}{d\alpha} (\alpha) \end{aligned}$$

$$\begin{aligned} &= \int_{\alpha}^{\alpha^2} \cos \alpha x dx + \frac{2 \sin \alpha^3}{\alpha} - \frac{\sin \alpha^2}{\alpha} \\ &= \frac{\sin \alpha x}{\alpha} \Big|_{\alpha}^{\alpha^2} + \frac{2 \sin \alpha^3}{\alpha} - \frac{\sin \alpha^2}{\alpha} \\ &= \frac{\sin \alpha^3}{\alpha} - \frac{\sin \alpha^2}{\alpha} + \frac{2 \sin \alpha^3}{\alpha} - \frac{\sin \alpha^2}{\alpha} = \frac{3 \sin \alpha^3}{\alpha} - \frac{2 \sin \alpha^2}{\alpha} \\ &= \frac{3 \sin \alpha^3 - 2 \sin \alpha^2}{\alpha} \quad // \end{aligned}$$

4. If  $\vec{r}$  is given as  $\vec{r} = (x^2 \sin y, z^2 \cos y, -xy^2)$  then compute  $d\vec{r}$

$$\frac{\partial \vec{r}}{\partial x} = (2x \sin y) \vec{i} - y^2 \vec{k} \quad \frac{\partial \vec{r}}{\partial y} = (x^2 \cos y) \vec{i} - (z^2 \sin y) \vec{j} - (2xy) \vec{k}$$

$$\frac{\partial \vec{r}}{\partial z} = (2z \cos y) \vec{j}$$

$$\begin{aligned} d\vec{r} &= \frac{\partial \vec{r}}{\partial x} dx + \frac{\partial \vec{r}}{\partial y} dy + \frac{\partial \vec{r}}{\partial z} dz \\ &= [(2x \sin y) \vec{i} - y^2 \vec{k}] dx + [(x^2 \cos y) \vec{i} - (z^2 \sin y) \vec{j} - (2xy) \vec{k}] dy \\ &\quad + [(2z \cos y) \vec{j}] dz \\ &= (2x \sin y dx + x^2 \cos y dy) \vec{i} + (2z \cos y dz - z^2 \sin y dy) \vec{j} \\ &\quad - (y^2 dx + 2xy dy) \vec{k} \quad // \end{aligned}$$

5.  $\phi = x^2 y z^3$   $\vec{A} = (xz, -y^2, 2x^2 y)$  are given in Cartesian coordinate system, compute;

i.  $\nabla \phi$  ii.  $\nabla \cdot \vec{A}$  iii.  $\text{div}(\phi \vec{A})$  iv.  $\text{rot}(\phi \vec{A})$

$$\begin{aligned} \text{i) } \nabla \phi &= \left( \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} \\ &= \frac{\partial (x^2 y z^3)}{\partial x} \vec{i} + \frac{\partial (x^2 y z^3)}{\partial y} \vec{j} + \frac{\partial (x^2 y z^3)}{\partial z} \vec{k} \\ &= 2xy z^3 \vec{i} + x^2 z^3 \vec{j} + 3x^2 y z^2 \vec{k} \end{aligned}$$

$$\begin{aligned} \text{ii) } \nabla \cdot \vec{A} &= \left( \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k} \right) \cdot (xz \vec{i} - y^2 \vec{j} + 2x^2 y \vec{k}) \\ &= \frac{\partial}{\partial x} (xz) + \frac{\partial}{\partial y} (-y^2) + \frac{\partial}{\partial z} (2x^2 y) = z - 2y \end{aligned}$$

$$\begin{aligned} \text{iii) } \text{div}(\phi \vec{A}) &= \nabla \cdot (\phi \vec{A}) = \nabla \cdot (x^3 y z^4 \vec{i} - x^2 y^3 z^3 \vec{j} + 2x^4 y^2 z^3 \vec{k}) \\ &= \frac{\partial}{\partial x} (x^3 y z^4) + \frac{\partial}{\partial y} (-x^2 y^3 z^3) + \frac{\partial}{\partial z} (2x^4 y^2 z^3) \\ &= 3x^2 y z^4 - 3x^2 y^2 z^3 + 6x^4 y^2 z^2 \end{aligned}$$

$$\begin{aligned} \text{iv) } \text{rot}(\phi \vec{A}) &= \nabla \times (\phi \vec{A}) = \nabla \times (x^3 y z^4 \vec{i} - x^2 y^3 z^3 \vec{j} + 2x^4 y^2 z^3 \vec{k}) \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 y z^4 & -x^2 y^3 z^3 & 2x^4 y^2 z^3 \end{vmatrix} \end{aligned}$$

$$\begin{aligned} [\phi \vec{A} &= (x^2 y z^3) (xz \vec{i} - y^2 \vec{j} + 2x^2 y \vec{k}) = (x^3 y z^4) \vec{i} - (x^2 y^3 z^3) \vec{j} + (2x^4 y^2 z^3) \vec{k}] \\ &= \left( \frac{\partial}{\partial y} (2x^4 y^2 z^3) - \frac{\partial}{\partial z} (-x^2 y^3 z^3) \right) \vec{i} + \left( \frac{\partial}{\partial z} (x^3 y z^4) - \frac{\partial}{\partial x} (2x^4 y^2 z^3) \right) \vec{j} \\ &\quad + \left( \frac{\partial}{\partial x} (-x^2 y^3 z^3) - \frac{\partial}{\partial y} (x^3 y z^4) \right) \vec{k} \\ &= (4x^4 y z^3 + 3x^2 y^3 z^2) \vec{i} + (4x^3 y z^3 - 8x^3 y^2 z^3) \vec{j} + (-2x y^3 z^3 - x^3 z^4) \vec{k} \end{aligned}$$

6. If  $\int_0^\pi \frac{dx}{\alpha - \cos x} = \frac{\pi}{\sqrt{\alpha^2 - 1}}$ , and  $\alpha > 1$  are given, using Leibnitz's

theorem compute

$$\int_0^\pi \frac{dx}{(2 - \cos x)^2}$$

$$\phi(\alpha) = \int_0^\pi \frac{dx}{\alpha - \cos x} = \pi (\alpha^2 - 1)^{-1/2}$$

$$\begin{aligned} \phi'(\alpha) &= - \int_0^\pi \frac{dx}{(\alpha - \cos x)^2} = -\frac{1}{2} \pi (\alpha^2 - 1)^{-3/2} \cdot 2\alpha \\ &= \frac{-\pi \alpha}{(\alpha^2 - 1)^{3/2}} \end{aligned}$$

$$\begin{aligned} \int_0^\pi \frac{dx}{(\alpha - \cos x)^2} &= \frac{\pi \alpha}{(\alpha^2 - 1)^{3/2}} \Rightarrow \int_0^\pi \frac{dx}{(2 - \cos x)^2} = \frac{2\pi}{(2^2 - 1)^{3/2}} \\ &= \frac{2\pi}{3^{3/2}} = \frac{2\pi}{3\sqrt{3}} \end{aligned}$$

7. Compute the area of the region bounded by the curve

$p = \alpha^2 \cos 2\phi$  on the lemniscate's on the xy plane.



$$\begin{aligned} 4 \int_{\phi=0}^{\pi/4} \int_{p=0}^{\alpha \sqrt{\cos 2\phi}} p \, dp \, d\phi &= 4 \int_{\phi=0}^{\pi/4} \left. \frac{p^2}{2} \right|_{p=0}^{\alpha \sqrt{\cos 2\phi}} d\phi \\ &= 2 \int_{\phi=0}^{\pi/4} \alpha^2 \cos 2\phi \, d\phi \\ &= \alpha^2 \sin 2\phi \Big|_{\phi=0}^{\pi/4} = \alpha^2 \sin \frac{\pi}{2} - \alpha^2 \sin 0 \\ &= \alpha^2 \end{aligned}$$